

AI-Based Research Paper Summarization and Categorization System

Problem Statement

Researchers, students, and professionals face significant challenges in keeping up with the rapidly growing volume of research papers published daily across multiple domains. Manually reading, analyzing, and extracting key insights from every paper is time-consuming and often impractical. This project aims to develop an AI-powered Research Paper Summarization and Categorization System that automatically fetches the latest research papers from RSS feeds, extracts essential information such as title, abstract, authors, and publication details, and generates concise, meaningful summaries using Large Language Models (LLMs). The system further identifies key technologies, highlights important contributions, and classifies papers into relevant research categories. By automating the research review process, the solution enables users to quickly understand emerging trends, discover valuable innovations, and save substantial time while staying updated with the latest developments in Artificial Intelligence, Machine Learning, Data Science, and related fields.

Objectives Automatically collect newly published research papers from RSS feeds. Generate concise AI-based summaries. Extract key technologies and contributions. Classify papers into research domains. Deliver daily research insights through workflow automation. **Technology Stack** n8n Workflow Automation RSS Feed Integration OpenRouter LLM Docker Prompt Engineering

Expected Outcome

An automated system that fetches, summarizes, categorizes, and presents research papers daily, reducing manual effort and improving research productivity.